

30. Morphological Operation using Dilation

AIM

To perform morphological dilation on an image using OpenCV.

PROCEDURE

1. Read the image.
2. Convert it to grayscale.
3. Create a morphological kernel.
4. Apply cv2.dilate().
5. Display the dilated image.

PROGRAM

```
import cv2
import numpy as np

# Read image
img = cv2.imread(r"C:\Users\chait\OneDrive\Documents\Desktop\cv lab image.jpg")

# Convert to grayscale
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# Create kernel
kernel = np.ones((5, 5), np.uint8)

# Apply dilation
dilated = cv2.dilate(gray, kernel, iterations=1)

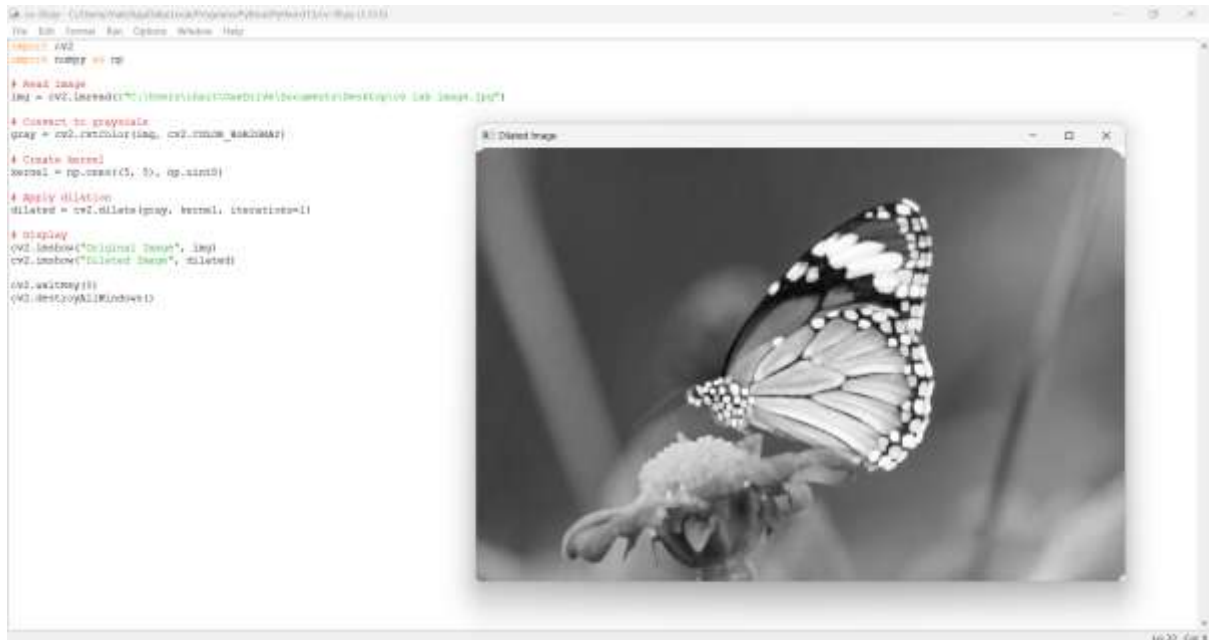
# Display
cv2.imshow("Original Image", img)
cv2.imshow("Dilated Image", dilated)
```

```
cv2.waitKey(0)
```

```
cv2.destroyAllWindows()
```

OUTPUT

The objects in the image become thicker after dilation.



Explanation

Dilation adds pixels around the boundaries of objects. It makes bright objects larger and can be used to fill small gaps and connect nearby regions.